Sangmyung Ha

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EDUCATION

Indiana University Bloomington Bloomington, IN
Ph.D. in Economics *Aug. 2020 -*
Sungkyunkwan University Seoul, Korea
B.A. in Economics *Mar. 2012 - Aug. 2019*

RESEARCH FIELDS AND INTERESTS

Econometrics, Macroeconometrics, Factor Models, Variational Inference

FELLOWSHIPS, HONORS AND AWARDS:

- 2023: IUB Economics Department, Class of 2020 Best Third Year Paper Award
- 2024 Fall-2025 Spring: Lloyd Orr Dissertation Fellowship

WORKING PAPERS

Estimation of Dynamic Factor Models in Time Domain
Abstract
We propose a novel approach for estimating dynamic factor models by introducing a *dynamic* dimension reduction technique in the time domain. This method is motivated by Brillinger's (1981) frequency domain dimension reduction method but is performed entirely in the time domain. The dynamic factor model considered in this paper is closely related to the generalized dynamic factor model presented by Forni et al. (2000). One of the defining characteristics of Forni et al. (2000)'s dynamic factor model is its double-sided filter of infinite length. Our estimation method is defined on the time domain, with a finite length filter. However, we show that if we increase the filter length to infinity at a suitable rate, we can estimate the dynamic factors consistently. Our method naturally handles a dynamic factor model with a finite length filter. In such a case, we show that a modified estimation method can estimate the dynamic factors at a similar rate as that of static factors estimated using principal component analysis.

Analysis of Global Yield Curve Using Functional Factor Model with Tensor Structure
with Yoosoon Chang (Indiana University) and Joon Y. Park (Indiana University)
Abstract
This paper introduces a functional factor model with tensor structure. Our model provides a framework to find and study common features existing in functional fluctuations over time and across crosssections with their effects having a tensor structure. Such common features, which are required to appear pervasively at all time and across all cross-sections, are defined formally as functional tensor factors in our model. We provide a computational algorithm to extract the functional tensor factors and their loadings within a general framework, as well as a method to estimate the number of them. Their consistency is established as we let the time span T and the cross-sectional dimension N increase simultaneously, under a very general set of conditions allowing for nonstationary as well as stationary functional time series. We apply our model and methodology to analyze fluctuations of the yield curves of major countries and find one global level factor and two global slope factors from which a curvature factor emerges for some countries.

PRESENTATIONS

- 33rd Annual Meeting of the Midwest Econometrics Group (MEG, Federal Reserve Bank of Cleveland), October 12-13, 2023
  -Paper title: *Estimation of Dynamic Factor Models in Time Domain*
- 18th International Symposium on Econometric Theory and Applications (SETA, Academia Sinica), May 29-30, 2024
  -Paper title: *Estimation of Dynamic Factor Models in Time Domain*
- 34th Annual Meeting of the Midwest Econometrics Group (MEG, University of Kentucky), November 1-2, 2024
  -Paper title: *Analysis of Global Yield Curve Using Functional Factor Model with Tensor Structure*, with Yoosoon Chang and Joon Park
- City St Gerge's, University of London Workshop on Applied Econometrics and Data Analysis, December 2024, 2024
  -Paper title: *Analysis of Global Yield Curve Using Functional Factor Model with Tensor Structure*, with Yoosoon Chang and Joon Park
- 18th International Symposium on Econometric Theory and Applications (SETA, University of Macau), June 1-3, 2025
  -Paper title: *Analysis of Global Yield Curve Using Functional Factor Model with Tensor Structure*, with Yoosoon Chang and Joon Park
- Advances in Empirical and Theoretical Econometrics (Sungkyunkwan University), August 16, 2025
  -Paper title: *Analysis of Global Yield Curve Using Functional Factor Model with Tensor Structure*, with Yoosoon Chang and Joon Park
- 35th Annual Meeting of the Midwest Econometrics Group (MEG, University of Illinois at Urbana-Champaign), October 4, 2025
  -Paper title: *Analyzing Global Economy using Tensor Factor Model*, with Yoosoon Chang and Joon Park

TEACHING/RESEARCH ASSISTANT

- **Associate Instructor (Full teaching responsibility)**
  -M512: Econometrics II (Graduate course) SP24
  -E370: Statistical Analysis For Business and Economics FA24
- **Graduate Assistant**
  -E370: Statistical Analysis for Business and Economics FA20
  -B251: Fundamentals of Econ for Business 1 SP21
- **Teaching Assistant**
  -E251: Fundamentals of Economics I FA22
  -B251: Fundamentals of Econ for Business 1 FA21
  -B252: Fundamentals of Econ for Business 2 SP22, SP23
  -E672: Macroeconometrics (Graduate course) FA22,FA23,FA24,FA25
  -E671: Econometrics III (Graduate course) FA23
  -E571: Econometrics I (Graduate course) FA25
- **Research Assistant**
  -**"Oil Price Volatility, Endogenous Regime Switching, and Inflation Anchoring"**
  Yoosoon Chang, Ana Maria, Herrera and Elena Pesavento, Forthcoming, Journal of Applied Econometrics
  -**"Unraveling Dynamic Interactions of Economic Activity and Climate Change"**

Yoosoon Chang, J. Isaac Miller, and Joon Y. Park

-**"How Do Macroaggregates and Income Distribution Interact Dynamically? A Novel Structural Mixed Autoregression with Aggregate and Functional Variables"**
Yoosoon Chang, Soyoung K, and Joon Y. Park

-**"Oil and the Stock Market Revisited: A functional VAR approach"**
Hilde C. Bjørnland, Yoosoon Chang, and Jamie Cross

SKILLS

Programming Languages
Matlab, Python, R, Julia

Languages
English (Fluent), Korean (Native)